§6-1 Areas Between Curves

Homework：1,4,7,14,19,24,29,44,45,47

Key

- $\sum$（離散的加法符號）

$\Updownarrow$

$\int$ （連續的加法符號）

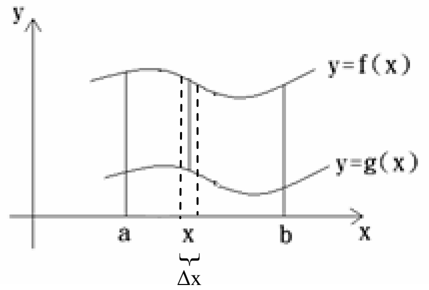
- A_n = **n** 維物體的體積

A_{n-1} =此物體 **n-1** 維截面的體積

$$\Rightarrow A_n = \int A_{n-1}dx = \int (截面)_{n-1}dx \approx \sum \underbrace{A_{n-1}}_{\text{n-1 維}} \underbrace{\Delta x}_{\text{1 維}}$$

例子：

(a)

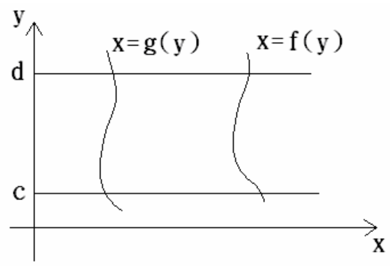


A_2 = Area between f and g from a to b

$A_1 = f(x) - g(x) = 1$維截面(線段)

$$A_2 = \int_a^b \left(f(x) - g(x) \right) dx$$

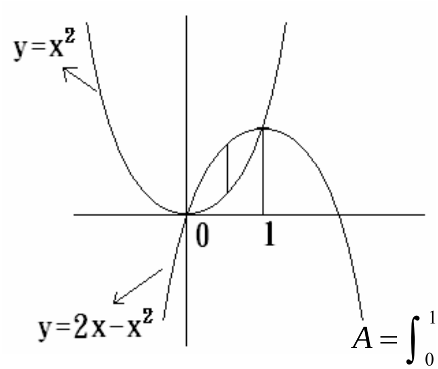
(b)



$$A_2 = \int_c^d A_1 dy = \int_c^d \left(f(y) - g(y) \right) dy$$

Example 1：Find the area of the region enclosed by the parabolas $y=x^2$ and $y=2x-x^2$.

Solution：



$$\begin{cases} y=x^2 \\ y=2x-x^2 \end{cases}$$

$$\Rightarrow x^2 = 2x - x^2$$

$$\Rightarrow x(x-1)=0$$

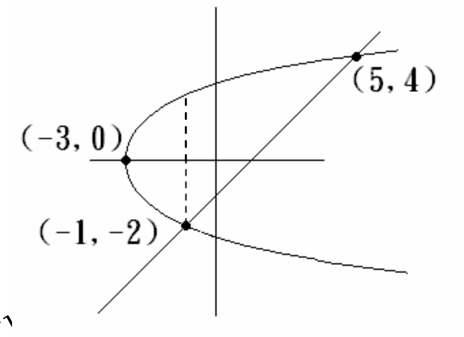
$$\Rightarrow \quad x=0 \quad or \quad 1$$

$$A=\int_0^1 \left(2x-x^2-x^2\right)\,dx = 2\int_0^1 \left(x-x^2\right)\,dx = 2\left(\frac{x}{2}-\frac{x^3}{3}\Big|_0^1\right)$$

$$=\frac{1}{3}.$$

Example 2：Find the area enclosed by the line $y=x-1$ and the parabola $y^2=2x+6$

Solution：



$$\begin{cases} y=x-1 \\ y^2=2x+6 \end{cases}$$

$$\Rightarrow (x-1)^2 = 2x+6$$

$$\Rightarrow \quad x=5 \quad or\ -1$$

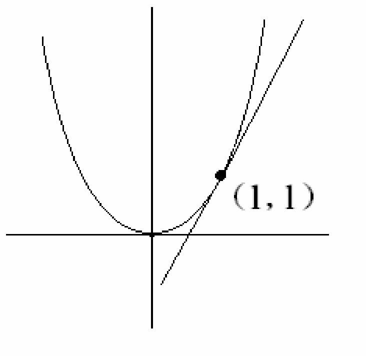
$$\Rightarrow \quad y=4 \quad or\ -2$$

$A=\int_{-3}^{-1}2$

$$=\int_{-2}^{4}\left[y+1-\frac{y^2-6}{2}\right]\,dy = 18.$$

Example 3：Find the area of the region bounded by $y=x^2$, the tangent line to this parabola at (1,1) , and the x-axis.

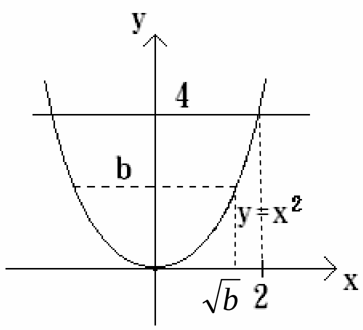
Solution：



$$y = x^2$$
$$\Rightarrow \quad y' = 2x$$
$$\Rightarrow y'(1) = 2$$
Tangent at $(1,1)$:
$$y - 1 = 2(x-1)$$
$$\Rightarrow \quad x = \frac{y+1}{2}$$
$$A = \int_0^1 \left(\frac{y+1}{2} - \sqrt{y} \right) dy = \frac{1}{12}.$$

Example 4：Find the number b such that the line y=b divides the region bounded by the curves $y=x^2$ and y=4 into two regions with equal area.

Solution：



$$\frac{1}{2}\int_0^2 \left(4 - x^2\right) dx = \int_0^{\sqrt{b}} \left(b - x^2\right) dx$$
$$\Rightarrow \frac{8}{3} = \frac{2}{3} b^{\frac{3}{2}}$$
$$\Rightarrow b = 4^{\frac{2}{3}}.$$